RES POLYTECHNIC, BANGALORE-91

CIE Theory Test I

Program	Mechanical Engineering	**Semester - III**
Course Name	Programming Concepts in Mechanical Engineering	**Test I**
Course Code	25ME34I	**Marks - 50**
Name of the Course Coordinator	Anushree B C	**Duration - 90 min**

Note: Answer any one full question from each section. Each question carries equal marks.

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	a) Draw a flowchart for an algorithm that finds the factorial of a given number using a loop. Show the input, initialisation, decision and output steps clearly.	L3	CO1	7
	b) Using the design of a simple beam-load calculator as an example, explain how dividing a problem into smaller sub-tasks makes an algorithm easier to write, test and debug.	L3	CO1	6
	c) Write Python statements that store a machine's name, its rated power in kW and whether it is currently running. Name the data type of each variable and justify your choice.	L3	CO1	6
	d) Explain how the input() function works in Python and what type of value it returns. Write a short program that reads the length and breadth of a steel plate from the user and displays its area.	L3	CO1	6
	OR			
2	a) Draw a flowchart to find the smallest of three temperature readings T1, T2 and T3.	L3	CO1	7
	b) Explain any four features of Python, including libraries such as NumPy and Matplotlib, that make it more suitable for engineering calculations than C++ or Java.	L3	CO1	6
	c) What is a Boolean value in Python? Explain with an example how it can be used to switch a cooling fan ON or OFF in a machine.	L3	CO1	6
	d) A program reads the diameter of a rod from the keyboard as text. Explain how type conversion is used to calculate the rod's cross-sectional area and write the Python code.	L3	CO1	6

Q. No	Questions	CL	CO	Marks
	SECTION II			
3	a) A lift is allowed to carry loads from 50 kg to 500 kg. Write a Python condition that reads the load and states whether it is permitted.	L3	CO1	7
	b) Explain the working of the break, continue and pass statements using a loop that scans a list of bolt diameters for defects. Give a short code snippet for each.	L3	CO1	6
	c) Explain function parameters and return values. Write a function that accepts the length, width and thickness of a plate and returns its volume, then call it for two different plates.	L3	CO1	6
	d) Distinguish between built-in and user-defined functions. Show with an example how a built-in function such as max() or sum() is simpler than writing your own.	L3	CO1	6
	OR			
4	a) A boiler temperature sensor gives readings that fall in a low, normal or high range. Write Python code that uses elif to print a suitable message for each range.	L3	CO1	7
	b) Compare the for loop and the while loop in Python. Give one situation in a mechanical workshop where a for loop is the better choice and one where a while loop is better.	L3	CO1	6
	c) Justify why you should write your own functions when a program repeats the same calculation many times, even though Python already provides many built-in functions.	L3	CO1	6
	d) Explain the scope of local and global variables using a program in which the value of g (acceleration due to gravity) is shared by several functions while intermediate results stay inside each function.	L3	CO1	6

Signature of the Course Coordinator

Signature of the HOD

Signature of the IQAC Chairman